Session 3 Course Note: Designing an AI Opportunity

For students in Innovation and Technology Management

Workshop connection: Session 3, 11:30–12:20

Purpose: Turn a broad interest in AI into a focused, measurable, and governable change to real work.

How to use this note

An AI opportunity is not a product name or a list of features. It is a hypothesis about changing a task, decision, or workflow for a defined user. This note helps you design that hypothesis before committing to a technology.

After reading, you should be able to:

- Describe the present workflow and problem without assuming that AI is the answer.
- Define the user, baseline, desired outcome, and consequence of error.
- Select a bounded AI role: assist, recommend, limited action, or no AI yet.
- Assess whether data and documents are suitable and permitted.
- Specify meaningful human control.
- Design the smallest useful test and a stop signal.
- Complete the workshop's AI Opportunity Canvas.

1. Begin with the work, not the technology

Weak proposal:

> *We should introduce a GenAI assistant because our competitors are using AI.*

Stronger starting point:

> *Field-service engineers spend substantial time locating the current procedure and relevant service history before diagnosing an unfamiliar fault. The information is distributed across several approved systems, and using an outdated document can create rework or safety risk.*

The second statement identifies a user, situation, workflow problem, information need, and consequence. It still does not assume that GenAI is the correct solution.

Before discussing AI, ask:

- What triggers the work?
- Who performs it?
- Which inputs are required?
- What steps occur today?
- Where do delay, error, duplication, or frustration arise?
- Which decisions require domain judgment?
- Who receives the output?
- What feedback shows that the work succeeded?

2. Map the current workflow

Use a simple sequence:

> ***Trigger → inputs → work steps → decision or output → recipient → feedback***

Example:

Element	Field-service example

Trigger	Customer reports abnormal vibration
Inputs	Asset ID, alarm log, service history, manuals, bulletins
Work steps	Confirm configuration, search documents, compare symptoms, inspect equipment
Decision/ output	Diagnostic hypothesis and next safe test
Recipient	Technician, service lead, customer contact
Feedback	Test result, repair outcome, repeat fault, updated service record

Mark four points on the workflow:

1. **Delay:** Where does work wait?
2. **Error:** Where does incorrect or incomplete information enter?
3. **Judgment:** Where is expertise required?
4. **Consequence:** Where could an error cause the most harm?

This map often reveals that the immediate need is better document ownership, a common asset identifier, a clearer approval process, or conventional search—not AI.

3. "No AI yet" is a valid design result

AI competes with other interventions:

- Remove an unnecessary step.
- Standardize terminology.
- Improve a form or checklist.
- Repair missing master data.
- Integrate two systems.
- Train users.
- Implement rules or conventional search.
- Clarify decision authority.

Use AI when its ability to infer, generalize, or generate offers meaningful advantage over simpler alternatives. A simpler solution may be easier to test, explain, maintain, and secure.

Choosing "no AI yet" is not resistance to innovation. It can be the most credible route toward later AI use because it improves the process and data first.

4. Write a problem statement before a solution statement

A useful problem statement contains:

- **User:** Who experiences the problem?
- **Situation:** When and where does it occur?
- **Present performance:** What happens now?
- **Consequence:** Why does it matter?
- **Boundary:** What is inside and outside the problem?

Template:

> *When **[situation]**, **[user]** currently **[workflow or difficulty]**, resulting in **[observable consequence]**. The problem includes **[scope]** and excludes **[scope]**.*

Avoid defining the problem as the absence of your preferred product. "We do not have a chatbot" is not a business problem.

5. Build a value hypothesis

An AI proposal is a hypothesis, not a promise.

Use:

> *For **[specific user]**, when **[situation]**, the proposed AI will **[defined role]**, which should improve **[measurable outcome]**, while the human remains responsible for **[decision or approval]**.*

Example:

> *For field-service engineers handling unfamiliar vibration alarms, a source-grounded assistant will retrieve and summarize the current approved procedures and matching service history, which should reduce median time to verified information while the engineer remains responsible for diagnosis and action.*

Notice what this does **not** promise. It does not claim that the assistant will diagnose the fault, prevent downtime, or improve safety unless those effects are separately tested.

Value needs a baseline

A claim of improvement requires present performance. Possible baselines include:

- Median search time per case.
- Percentage of cases using the current document version.
- First-time-right rate.
- Rework hours.
- Mean time to acknowledge or resolve an issue.
- On-time delivery.
- Unplanned downtime.
- User or customer experience.
- Number and severity of quality or safety incidents.

If no baseline exists, the first part of the pilot may be measurement rather than AI deployment.

6. Define the AI role and autonomy

Use the lowest level of autonomy that can test the value hypothesis.

Role	What the AI may do	What the human must do	Suitable first-pilot example
Assist	Retrieve, summarize, draft, classify	Check, edit, and complete	Draft a service brief from approved documents
Recommend	Compare options and propose a choice	Review evidence and make the decision	Recommend which inspection to prioritize
Limited action	Perform a narrow, reversible, pre-approved action	Monitor exceptions and retain stop authority	Create a draft work order within defined rules
No AI yet	No AI role	Improve process, rules, data, or integration	Standardize document metadata first

Do not use "human in the loop" as a complete control. State:

- Which human role reviews.
- At what point review occurs.
- What evidence is visible.
- Which checks are mandatory.
- Whether the reviewer has time and competence.
- Whether the reviewer can reject or override.

- Who handles exceptions and incidents.
- Who remains accountable.

A reviewer who lacks evidence, authority, or time is not an effective control.

7. Data readiness is part of opportunity design

AI cannot repair every data problem. Sometimes it makes poor data easier to consume and therefore more dangerous.

Review six dimensions.

Permission

- Who owns the documents or data?
- May they be used for this purpose and in this environment?
- Are personal, confidential, export-controlled, or licensed materials involved?

Relevance

- Do the data describe the users, equipment, locations, languages, and situations in scope?
- Are rare but important cases represented?

Quality

- Are records accurate, complete, consistently structured, and deduplicated?
- Do labels represent the intended concept?

Currency and configuration

- Which version is authoritative?
- Can the system distinguish obsolete procedures from current ones?
- Does it know the asset configuration and effective date?

Traceability

- Can an output be connected to the source, record, model version, and time?
- Can a reviewer inspect the original evidence?

Security and robustness

- Could a malicious or incorrect document manipulate the output?
- Are permissions preserved during retrieval?
- What happens if the service or data source is unavailable?

The FFI/Navy HR assistant illustrates the value of a purpose-specific source collection, while the DNV/Green Cargo case illustrates how a focused classification task depends on clear categories and continuous attention to data quality (FFI (https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-felteksperiment-med-digital-hr-assistent); DNV (https://www.dnv.com/cases/green-cargo-uses-ai-to-improve-data-quality-and-efficiency-245284/)). Neither example proves that your data are ready.

8. Separate feasibility, value, usability, and acceptability

An AI opportunity can succeed technically and still fail organizationally.

Dimension	Question

Technical feasibility	Can the system perform the task under real operating conditions?
Operational fit	Can it be integrated into the workflow, systems, roles, and response times?
User value	Does it solve a meaningful problem for the intended user?
Business or public value	Does the improvement matter to cost, quality, safety, delivery, service, or mission?
Legal and ethical acceptability	Is the purpose and method permitted and defensible?
Organizational readiness	Are ownership, skills, support, incentives, and change capacity present?

Do not use a successful technical demonstration as evidence for all six dimensions.

9. Design measures before the pilot

Measures shape behaviour. Choose a small set that represents both value and harm.

Value measure

Examples:

- Time to locate verified information.
- Rework per completed task.
- First-time-right rate.
- Unplanned downtime.
- Response time.
- On-time completion.
- User-rated usefulness with a defined scale.

Quality measure

- Supported-claim rate.
- Classification precision and recall.
- Percentage of outputs using the correct document version.
- Expert-rated completeness.

Risk or guardrail measure

- Confidential-data incidents.
- Unsupported safety recommendations.
- Performance differences across groups, languages, or sites.
- User overrides, complaints, and escalations.
- Cases in which the human could not verify the evidence.

Adoption or process measure

- Eligible users who try the tool.
- Repeat use for the intended task.
- Completion of required verification steps.

Usage alone is not value, but it may help explain why value did or did not appear.

10. The smallest useful test

A useful test is large enough to produce evidence and small enough to stop safely.

Define:

- A limited user group.
- One task and workflow.
- One approved source set or dataset.
- A defined period.
- Clear exclusions.
- A comparison with current practice.
- Value, quality, and risk measures.
- A named owner.
- A stop signal.

Example:

> *For six weeks, ten field-service engineers use a source-grounded assistant only to find and summarize approved procedures for two equipment families. The assistant cannot diagnose faults or create work orders. Half of eligible cases are handled using the current process for comparison. We measure search time, correct-document rate, unsupported claims, and user overrides. The service lead pauses the test after any untraceable safety-critical recommendation or unauthorized document access.*

This is more informative than a general invitation to "try the chatbot."

11. Stop signals and reversibility

A stop signal is decided before enthusiasm or sunk cost changes judgment.

Possible signals include:

- Personal or confidential data appear outside the permitted environment.
- A safety-related recommendation cannot be traced to approved evidence.
- Error rates exceed an agreed threshold.
- Performance differs materially by language, group, product, or location.
- Users cannot identify incorrect output reliably.
- Review work costs more than the time saved.
- The system changes a record or sends information without required approval.

Stopping, narrowing, or redesigning a pilot is a valid learning result.

12. Human and organizational change

The AI opportunity changes more than task time. Consider:

- Which expertise becomes more or less visible.
- Whether junior employees lose learning opportunities.
- Whether users become over-reliant on recommendations.
- Whether employees fear monitoring or displacement.
- Whether new errors move to another department.
- Who handles feedback and updates.
- Whether suppliers gain access to data or operational dependence.
- Whether the process works across languages and cultures.

Engage the people who perform and receive the work. They understand exceptions that process diagrams often omit.

13. A completed canvas example

The workshop uses the Slim AI Opportunity Canvas (the A3 worksheet from the workshop materials). The worked example below is primarily for **post-workshop study**. If your team selects the field-service knowledge challenge and reads this beforehand, do not reproduce it. Find your own problem boundary, evidence needs, measures, safeguards, and test design; the example is one possible interpretation, not a model answer.

1. User, problem, and baseline

Field-service engineers handling unfamiliar vibration alarms. Current median verified-document search time is not known; a two-week baseline study is required. Wrong document versions can cause rework and safety risk.

2. AI role

Assist. Retrieve and summarize approved procedures and matching service history. Do not diagnose, recommend an equipment change, contact the customer, or create a work order.

3. Permitted sources and exclusions

Approved manuals, bulletins, and de-identified service histories for two equipment families. Exclude customer personal data, commercial correspondence, draft engineering documents, and all other products.

4. Value and measure

Reduce median time to verified information and increase correct-document-version use. First establish the baseline.

5. Human owner and check

The engineer checks document ID, revision, equipment configuration, and supporting passage. The service lead owns the pilot and handles incidents.

6. Smallest test and stop signal

Ten engineers, two equipment families, six weeks. Stop after unauthorized source access or any untraceable safety-critical statement.

The example is deliberately narrow. It can expand only after evidence supports expansion.

14. Common design mistakes

Technology-led problem framing

“Use GenAI in maintenance” is too broad to test.

Vague value

“Improve efficiency” does not define a measure or baseline.

Generic oversight

“A human checks” does not identify the role, evidence, competence, time, or authority.

Assumed data readiness

Available documents may be outdated, inconsistent, restricted, or unrepresentative.

Maximum autonomy as the goal

More autonomy increases possible impact and control requirements. It is not automatically more valuable.

Pilot without a comparison

If you do not measure current practice, you may confuse novelty or selection effects with improvement.

Success without a stop condition

A pilot that can only be declared successful is marketing, not an experiment.

15. Apply this note to Nordic Systems

For your assigned challenge:

1. Map the current workflow.
2. Mark delay, error, judgment, and consequence.
3. Identify one non-AI improvement.
4. Write the problem statement.
5. Select the lowest useful AI role.
6. Name permitted sources and explicit exclusions.
7. Choose one value measure and one risk measure.
8. Name a real human role and mandatory check.
9. Define the smallest test and one stop signal.
10. Complete the AI Opportunity Canvas.

16. Reflection questions

1. Which industrial problems are poorly suited to GenAI but well suited to prediction, classification, optimization, or no AI?
2. When could a source-grounded assistant still present the wrong document?
3. How might a pilot improve measured task time while making the wider workflow worse?
4. What evidence would persuade you to move from assist to recommend? From recommend to limited action?
5. Which groups benefit from your proposal, and which groups carry the risk?

Key terms

- **Use case:** A defined application of a capability for a specific user, task, and context.
- **Value hypothesis:** A testable statement connecting a user, intervention, and measurable improvement.
- **Baseline:** Present performance before the intervention.
- **Guardrail measure:** A measure used to detect unacceptable harm or degradation.
- **Autonomy:** The degree to which a system selects or performs actions without immediate human approval.
- **Data readiness:** The suitability, permission, quality, relevance, and governance of data for the proposed purpose.
- **Stop signal:** A predefined observation or incident that pauses or ends a test.
- **Reversibility:** The ease with which an intervention or its effects can be undone.

References and further reading

- NIST. AI Risk Management Framework (https://www.nist.gov/itl/ai-risk-management-framework) (2023; revision work ongoing in 2026).
- NIST. AI RMF Playbook and Resource Center (https://airc.nist.gov/).
- OECD. Explanatory memorandum on the updated definition of an AI system (https://oecd.ai/en/ai-publications/explanatory-memorandum-on-the-updated-oecd-definition-of-an-ai-system) (2024).
- FFI. Økt produktivitet med kunstig intelligens? (https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-felteksperiment-med-digital-hr-assistent), report 25/008 (2025).
- DNV. Green Cargo uses AI to improve data quality and efficiency (https://www.dnv.com/cases/green-cargo-

uses-ai-to-improve-data-quality-and-efficiency-245284/).

- Datatilsynet. Recommendations for purchasing and using AI-based systems (https://www.datatilsynet.no/regelverk-og-verktoy/rapporter-og-utredninger/kunstig-intelligens/anbefalinger/).

Session takeaway

The best first AI opportunity is usually narrow enough to test, valuable enough to matter, and safe enough to stop.